Preliminaries of Statistical Inference

Benjamin Abshire[†]
Indiana University Bloomington

October 15, 2026

I A General Setting

Consider that we have n numbers: $x_1, ..., x_n$, which are believed to be randomly generated in the way that they take on different values. One mechanism could be a toss of a coin. We call this set of numbers *data*. We can suggest that there are realized values of these n random variables $X_1, ..., X_n$ defined on a common probability space $(\Omega, \mathcal{F}, P)$. Recall that a probability space is a structure where we can specify possible outcomes, the events formed from those outcomes, and the probability assigned to said events. In other words,

$\Omega =$ what can happen, $\quad \mathcal{F} =$ events we can form from Ω, $\quad P =$ how likely those events are

Since these random variables are defined on a common probability space, we can assume that any outcomes ω in the sample space has taken value $x_1, ..., x_n$ that we can observe. Let $X = (X_1, ..., X_n)'$. Then

$$X(\omega) = \begin{pmatrix} X_1(\omega) \\ \vdots \\ X_n(\omega) \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Statistical inference is the study of the distribution of the random vector X. Equivalently, the study of the joint distribution $X_1, ..., X_n$. The distribution of X can be assumed to belong to a certain family of distributions, denoted by

$$\mathcal{P} = \{P_\theta | \theta \in \Theta\}$$

Where θ is an index called the parameter and Θ is the parameter set. Each member P_θ of the family $\mathcal{P}$ is indexed by θ. For example, one's student ID number is a parameter in the set of the IU community. We can formulate the family in terms of the density p_θ. Recall that a density is a function that lets us obtain probabilities by integration. For a continuous random variable X and any set of values A,

$$P(X \in A) = \int_A p(x)dx$$

[†]Material from *Advanced Probability and Statistics for Economists* by Yoosoon Chang and Joon Y. Park and ECON E-571: Econometrics I instructed by Dr. Joon Y. Park.

Statistical inference with a parametric family of distributions is called the **parametric approach**. If we do not restrict to a certain family, then we follow the **nonparametric approach**.

a. Exercises

1. Suppose we want to find the distribution of $X = (X_1, ..., X_n)'$ using an observation $x = (x_1, ..., x_n)'$. Generally, this does not make sense. Explain why

Solution. Generally, this does not make sense because we have no restrictions on the families of distributions we are considering. As such, there could be infinitely many joint distributions that could have produced the one observation x, making it impossible to use it to find the distribution of X.

2. Would it be more sensible to try to find the distribution of $X = (X_1, ..., X_n)'$ using an observation $x = (x_1, ..., x_n)'$ if $X_1, ..., X_n$ are i.i.d? How about if we know that $X_1, ..., X_n$ are i.i.d $\mathbb{N}(\mu, \sigma^2)$?

Solution. Yes, it is more sensible if $X_1, .., X_n$ are i.i.d since all observations provide the same information about the underlying distribution. It is also sensible if $X_1, ..., X_n$ are i.i.d $\mathbb{N}(\mu, \sigma^2)$ because we now have a restriction to the normal family and only need to estimate μ, σ^2.

II Statistical Inference

The family of distributions $\mathcal{P}$ is thought to contain a member that is the true distribution P_{θ_0}. So, the study of the distribution of X is just choosing that member P_{θ_0}. Let

$$X : \Omega \to \mathcal{X}$$

Where $\mathcal{X}$ (which is $\mathbb{R}^n$ in most cases) is called the state space. A state space is the set of all possible values a random variable or vector can take on. So, statistical inference is preformed by constructing a function

$$\tau : \mathcal{X} \to \mathcal{T}$$

Where $\Theta \subset \mathcal{T}$. Such a function τ or its corresponding random element $T = \tau(X)$ is called a **statistic**. There are two types of statistics depending on what we want to achieve. In statistical inference there are two common procedures: **estimation** and **hypothesis testing** on θ. If a statistic or its random element are used to estimate θ, then it is called an **estimator**. The realized value $\tau(x)$ of T is called an **estimate**. An estimator or estimate for the parameter θ is denoted by $\hat{\theta}$. If used for hypothesis testing, (where we partition Θ into two pieces and select one depending on the value of τ) then the they are called a **test statistic**. To visualize statistical inference, consider the following figure:

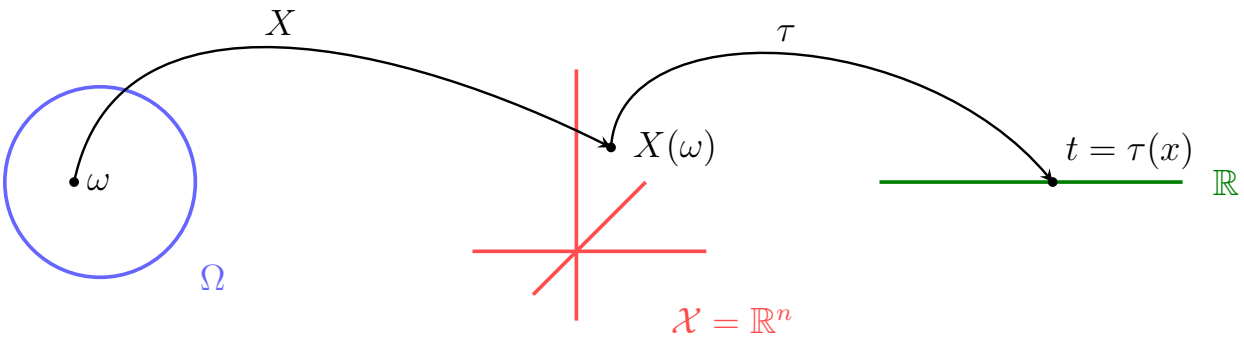


Figure 1: A statistic as a composition of maps: an outcome $\omega \in \Omega$ is mapped by the random vector $X : \Omega \to \mathcal{X}$ to the data $X(\omega) = x$, which the statistic $\tau : \mathcal{X} \to \mathcal{T}$ maps to $t = \tau(x)$.

III Sufficiency

Let $T = \tau(X)$ be a statistic and $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$ be a family of distributions of X. We can say that T (or τ) is **sufficient** for $\mathcal{P}$ (or for θ) if the conditional distribution of X given $T = t$ does not depend on θ for all t. If T is a sufficient statistic, the conditional distribution of X given $T = t$ is uniquely determined irrespective of θ. If it is not sufficient, the conditional distribution of X given $T = t$ would depend on θ. Note this is the case because the distribution of X is unknown, it can be any member of the family $\mathcal{P}$. Another way of stating sufficiency: on any member of the partition made by T, the distribution of X is the same across all θ.

If T is sufficient, then we can write

$$p_\theta(x, t) = p(x|t)p_\theta(t)$$

Where $p_\theta(x, t)$ is the joint density of the random variable X. Note that t has all the information about θ. Hence, $p(x|t)$ has no subscript θ, since it does not depend on it.

a. Exercises

1. Suppose $X =_d \mathbb{N}(0, \sigma^2)$ and $T = |X|$. Is statistic T sufficient?

Solution. Yes. For $T = t$, X can only take on two values: $t, -t$. These are equally likely since the distribution of X is symmetric around the origin. So, the conditional distribution of X given $T = t$ is concentrated around two points $t, -t$ which both have probability $1/2$. This does not change if σ^2 changes. Therefore T is sufficient.

2. True or false: If T is sufficient, then is any function of T sufficient? Prove or provide a counter example

Solution. False. Let $T = X$ and $X =_d \mathbb{N}(\mu, 1)$. Then, let $S = f(T) = 0$. S is therefore constant regardless of T. If S sufficient, then the conditional distribution $X|S = 0$ by definition doesn't depend on μ. But, $X|S = 0 \sim \mathbb{N}(\mu, 1)$ depends on μ so it is not sufficient. Therefore, if T is sufficient then it is not true that any function of T is sufficient.

IV Bayesian Approach

Consider the random experiment of tossing a coin. The **classical** or **frequentists'** approach assumes that there is an unknown fixed number between 0 and 1 that represents the probability of the coin landing on heads. It is defined as the limit of the proportion of the coin landing on heads if we toss it infinitely many times. This is the approach we have been studying.

However, there is another approach called the **Bayesian** approach. In the same coin example, the probability is regarded as a random variable, the distribution of which is to be found. The unknown parameter is a random variable Θ, and the conditional distribution of X given $\Theta = \theta$ is a family of densities $p_\theta(x)$ that we use to model the distribution of X. Note that the density $p_\theta(x)$ of X is now written as $p(x|\theta)$. The density $p(\theta)$ of Θ reflects our subjective belief on the true parameter value. It is called the prior density. The prior belief is modified according to the observation of x following the Bayes rule:

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

Where $p(x) = \int p(x|\theta)p(\theta)d\theta$. Again note that we can think of $p(x|\theta)$ as the statistical model. The density $p(\theta|x)$ is called the posterior density i.e, what is believed after we observe the data and apply Bayes rule. The goal of Bayesian analysis is to obtain the posterior density $\Big(p(\theta|x)\Big)$ from a prior density $\Big(p(\theta)\Big)$ using a statistical model $\Big(p(x|\theta)\Big)$. Explained another way: Before we observe the data

- $\Theta \sim p(\theta), X|\Theta = \theta \sim p(x|\theta)$
  - i.e. Θ, X are random

After we observe $X = x$

- x is known so only Θ is unknown
- Given $X = x$ what do we believe about Θ?
  - $\Theta|X = x \sim p(\theta|x)$
- i.e. we want to obtain the posterior distribution from the prior and the model.

In Bayesian analysis, we write

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

i.e. the posterior density is **proportional** to the model times prior density. We can say this since $p(x)$ is just a normalizing constant to make $\int p(\theta|x)d\theta = 1$. A useful table to distinguish between the two approaches follows:

	Frequentist	Bayesian
θ	fixed, unknown	random variable Θ
X	random	random conditional on Θ
Goal	estimate/test θ	learn distribution of $\Theta \mid X = x$

a. Exercises

1. In Bayesian analysis, we often use an improper prior and let the distribution of θ be uniform over the entire real line. Explain how the Bayesian approach is compared with the classical approach in this case.

Solution. Let $p(\theta) \propto 1$, i.e. before seeing the data we do not favor any value of θ over another. By Bayes rule, $p(\theta|x) \propto p(x|\theta)p(\theta) \implies p(\theta|x) \propto p(x|\theta) \cdot 1$. So we have that $p(\theta|x) \propto p(x|\theta)$. Recall that the classical maximum likelihood estimator (MLE) is $\hat{\theta}_{ML} = \arg\max_\theta p(x|\theta)$ So, maximizing the posterior is equal to maximizing the likelihood, i.e.

$$\arg\max_\theta p(\theta|x) = \arg\max_\theta p(x|\theta)$$

So, in this case the two approaches are very similar.